

Evolution Strategies, Covariance Adaptation and CMA-ES

Optimization Team

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September 6, 2026

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SEAMS School 2026: Applying Mathematics to Real-World Applications

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Introduction

The objective of the study is to understand the progression

random search → evolution strategy → adaptive Gaussian search → CMA-ES

After: we shall see how important it is to adapt a full covariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

Derivative-Free Optimization

Let $D \subset \mathbb{R}^n$ and $f : D \rightarrow \mathbb{R}$. We want to minimize the function f

$$\min_{x \in D} f(x) \tag{1}$$

If f is smooth (i.e., continuously differentiable) then to solve (1), we can use gradient descent methods, such as steepest descent method or Newton-like methods, which can generally generate the solution by updating at each iterate

$$x_{n+1} = x_n - \eta \nabla f(x_n)$$

We are here to answer three questions:

1. What information does a gradient use that a population-based methods does not have directly?

An Gradient methods use the first-order derivative of the objective function. The gradient provides the descent direction in the search process to obtain the minimizer which the population-based methods do not access. Derivative-free methods evaluate the objective function values at the a set of candidate points and select the best that improves the function value. These methods however, help us to do optimization with nonsmooth functions which is practical in real-world applications.

2. Why can derivative-free methods still recover preferred directions of descent?

An Yes, DFO can recover the descent direction even without gradient information. To obtain so, DFO implicitly approximate the gradient-like information using te function values and they structure their search so that at least one trial direction is guaranteed to be descent direction.

3. What difficulties arise when d is large?

